Reusable Components from Existing Research

This document summarizes the components from the previous research that can be directly reused in the proposed Agentic AI Traffic Management System.

1. Congestion Formula

Average Speed (km/h) = (Distance / Travel Time) × 3600

Congestion Index = (Free Flow Speed – Average Speed) / Free Flow Speed

Inputs

- Travel Time
- Distance
- Free Flow Speed

Outputs

- Congestion Index

2. Emission Formula

CO■ (g/km) = $a + bV + cV^2$

Where: $a = 300$, $b = -5$, $c = 0.1$, V = Average Speed

Inputs

- Average Speed

Outputs

- CO■ Emission per km
- CO■ Emission per Trip

3. Decoupling Formula

Δ Congestion = (Congestion Index – Baseline Congestion) / Baseline Congestion

Δ Emission = (Emission – Baseline Emission) / Baseline Emission

Decoupling Index = Δ Emission / Δ Congestion

Inputs

- Congestion Index
- Emission Values

Outputs

- Coupled / Decoupled Classification

4. XGBoost Features

- average_speed_kmh

- congestion_index
- total_delay_sec
- num_intersections
- distance_km
- intersection_type
- network_type
- intersection_type_Signalized
- intersection_type_Urban_Mixed
- network_type_Grid_Signalized
- network_type_Real_DC

5. XGBoost Output

Target Variable: decoupled_label

1 = Decoupled

0 = Not Decoupled

Model: XGBoost Classifier

Purpose: Predict the probability that a traffic scenario exhibits emission–congestion decoupling.

6. Traffic Types Studied

- Signalized Intersections – Controlled by traffic lights with moderate delays.
- Unsignalized Intersections – No traffic signals and generally lower delays.
- High-Volume Intersections – Heavy traffic demand with high congestion.
- Real Washington DC Dataset – Real-world travel-time observations used for validation.

7. Mapping to the New Agentic AI Project

- Congestion Formula → Congestion Agent
- Emission Formula → Emission Agent
- Decoupling Formula → Decoupling Agent
- XGBoost Model → Prediction Agent
- New Components: Propagation Agent, Decision Agent, Action Agent